Symposium “Recent Advances in Meta-Analysis” · 28–29 May 2026 · Göttingen

Mortality by COPD Severity (GOLD 1–4): A Meta-Analytic Workflow for Deriving German 1-Year Risks for Decision Models

QR code
Poster / details

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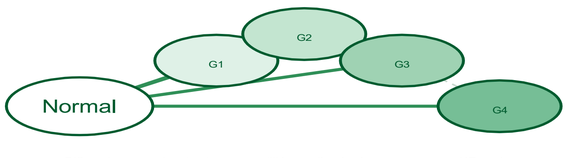
Why this matters

Decision models for COPD need credible mortality inputs by disease severity. The key contribution here is methodological: a transparent bridge from heterogeneous cohort evidence to model-ready parameters.

- Supports long-term cost/effect simulation
- Makes assumptions explicit
- Systematic search in two databases (PubMed + Europe PMC)
- Clearly defined in- and exclusion criteria
- Avoids ad hoc selection of single cohorts

Why network meta-analysis?

Included cohorts did not all use the same comparator. Network meta-analysis (NMA) places GOLD grades on one common HR scale and combines evidence across population-based and clinical settings.



Different comparator structures can still be synthesized on one common HR scale

Evidence base

34	1.46M	2.2–30 y
studies	participants	follow-up

Eligibility: English-language cohorts reporting HRs by GOLD grade, at least 1 year of follow-up, and adjustment for age and sex (+ smoking in population-based studies).

Risk of bias assessed with the Newcastle–Ottawa Scale.

Methodological workflow

1. Find the evidence

Searched PubMed + Europe PMC

- Backward + forward citation search
- Authors contacted for major cohort data, obtained data from 5 more large studies
- Focus on adjusted HRs by GOLD grade

2. Harmonize and assess

- Require age/sex adjustment
- Require smoking adjustment in population samples
- Check follow-up and reporting quality
- Newcastle–Ottawa risk-of-bias assessment

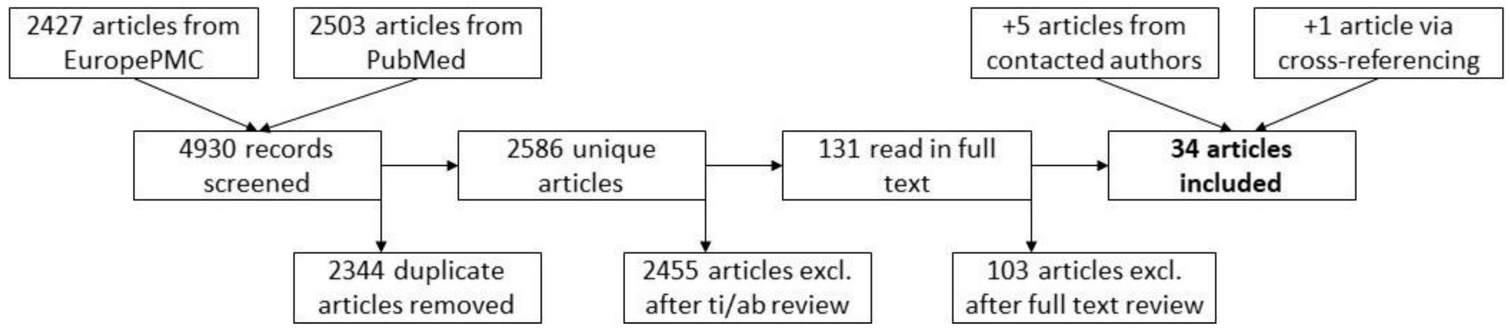
3. Synthesize the network

- Run network meta-analysis in R (netmeta)
- Combine studies with different comparators
- Estimate pooled HRs for GOLD 1–4 on a common scale
- Examine subgroup/sensitivity patterns

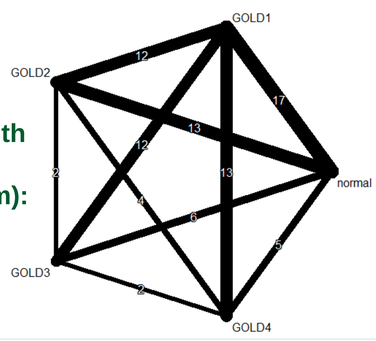
4. Translate to model inputs

- Combine pooled HRs with German life-table mortality
- Derive 1-year risks by GOLD grade, age group, and sex
- Pass the estimates into natural-course Markov models

Flow diagram of study selection:



Network plot with N(studies) per comparison/arm):



General structure of a typical COPD Markov model



The resulting inputs can be varied transparently in scenario and sensitivity analyses.

Key **assumptions** of network meta-analysis:

- Transitivity: effect modifiers compared across contrasts.
- Common scale: NMA re-expresses HRs against normal spirometry.

Robustness checks:

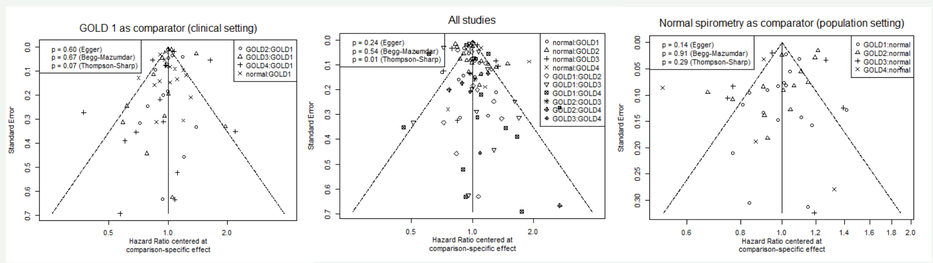
- Small-study effects (funnel plots, see above)
- Heterogeneity + inconsistency
- Leave-one-out sensitivity analysis
- Stratification by follow-up duration
- Stratification by setting (population vs. clinical)
- Excluded studies adjusting for potentially causal variables

Results in brief

Pooled mortality **hazard ratios** (reference: normal spirometry). Monotonic ordering across GOLD grades; the same ranking was observed in population-based and clinical settings.

GOLD 1	1.09
GOLD 2	1.57
GOLD 3	2.40
GOLD 4	4.50

Funnel plots to visualize small-study effects (potential publication bias):



Further key **results**:

- Stronger associations in shorter-follow-up studies (<9 years)
- Funnel plots largely symmetric, suggesting no strong small-study effects

Decision-analytic use case

Why the workflow is useful for decision models:

The synthesis feeds the natural course of COPD in a Markov model. What matters is not only that mortality rises monotonically, but that the final input is stratified by severity, age, and sex and is therefore suitable for scenario analysis.

Examples of derived German **1-year mortality** risks:

Female, GOLD 1, age 45–49: **0.15%**

Female, GOLD 2, age 65–69: **1.59%**

Male, GOLD 3, age 75–79: **10.1%**

Male, GOLD 4, age 80–84: **28.7%**

Take-home message Network meta-analysis harmonized heterogeneous COPD cohorts and made it possible to derive transparent, German-specific 1-year mortality inputs for Markov decision models.

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Funded by the German Federal Ministry of Education and Research (BMBF), grant 01EK2203C.